

Naveenkumar Narsozhan

Los Angeles, California — 213-796-2976 — narsozha@usc.edu — LinkedIn/naveenkumar — github.com/Naveenkumar0105

EDUCATION

University of Southern California, *MS in Computer Science* – **CGPA: 3.82/4.0** Aug 2025 – May 2027

- Relevant Courses: Analysis of Algorithms, Information Retrieval, Machine Learning, Deep Learning

NIT Puducherry, *B.Tech in Computer Science and Engineering* – **CGPA: 9.3/10.0** Dec 2020 – May 2024

- Relevant Courses: Artificial Intelligence and Expert Systems, Soft Computing Techniques, Pattern Recognition, Big Data

TECHNICAL SKILLS

Languages: Python, C, C++, SQL, JavaScript, TypeScript, Bash

Frameworks: PyTorch, TensorFlow, Sklearn, HuggingFace, YOLO, OpenCV, LangChain, LlamaIndex, FastAPI

Domains: ML/DL, NLP, CV, Agentic AI, RAG, LLMs, Prompt Engineering, Query Optimization

Tools & Infra: Azure, Docker, Git, PostgreSQL, MongoDB, ChromaDB, Vercel, Supabase, Linux, CI/CD

PROFESSIONAL EXPERIENCE

Microchip Technology Inc. - Roseville, CA, USA, *Intern* May 2026 – Aug 2026

- Engineered a cross-platform Python GUI to interface with custom SSD firmware for Future of Memory and Storage (FMS) 2026, driving technical demonstrations of two storage technologies to 500+ clients.
- Automated PCIe/NVMe hardware benchmarking for AI workloads by writing Python scripts to stress-test evaluation boards across 1000+ iterations, identifying firmware bottlenecks and debugging complex customer edge cases.

HCL Tech - Madurai, Tamil Nadu, India, *Technical Lead* Sep 2024 – Jul 2025

- Trained in Microsoft Azure services, including App Service, Azure SQL Database, and Azure DevOps, gaining expertise in cloud migration and deployment pipelines.
- Contributed to migrating a legacy corporate website (50K+ monthly users) from on-premise servers to the Azure platform, improving availability and reducing infrastructure costs by an estimated 20%.

National Institute of Ocean Technology - Chennai, India, *ML Research Intern* May 2023 – Jun 2023

- Performed extensive exploratory data analysis (EDA) on 1M+ meteorological, oceanographic, and ship trajectory data points to identify key environmental factors impacting fuel efficiency and optimal navigation.
- Developed a robust data preprocessing pipeline to clean, integrate, and normalize large raw datasets (CSV-to-Excel transformation, missing value handling, feature scaling) to enable reliable downstream analysis.

PROJECTS

VibeArchitect (Agentic Content Pipeline) | *Special Mention, TrojanHacks USC* Feb 2026

- Architected an agentic AI pipeline using Google Gemini API and HuggingFace (FLUX.1) to automate content trend analysis and thumbnail generation, winning the Special Mention (Solo Hacker) award at ACM TrojanHacks.
- Engineered a full-stack React and Node.js/Express application, integrating the Presaige API for automated visual auditing and implementing custom long-polling workflows to maintain 100% system uptime.

Dynamic Vision Inference Pipeline | *PyTorch, YOLOv11, Object Detection* Jan 2026 – Apr 2026

- Architected a collaborative computer vision pipeline for X-ray security screening, designing a dynamic routing engine to cascade inference between a lightweight NanoDet-Plus model and a high-capacity YOLOv11L detector.
- Optimized inference efficiency across 36,800+ COCO-format test images by implementing confidence-threshold routing, reducing total processing time by 54% while improving baseline AP@50 accuracy by 6.9%.

Kuri AI – LLM-Driven Task Understanding System | *Multimodal LLMs, Agentic AI* Nov 2025 – Present

- Architected an LLM-powered task intelligence pipeline using the Gemini API for open-set intent classification, semantic parsing, hierarchical task decomposition, and structured entity extraction from multimodal (voice + text) inputs.
- Implemented an agent-based orchestration framework with API-driven Google Calendar integration, designed RESTful backend services with Supabase, and deployed the production system on Vercel with serverless functions.

Valorant Strategy Assistant - RAG Chatbot | *Semantic Search, ChromaDB, NLP* Sep 2025 – Dec 2025

- Built a Retrieval-Augmented Generation (RAG) system using sentence-transformers and ChromaDB, indexing 50+ Valorant knowledge documents for low-latency semantic search.
- Integrated a quantized 7B parameter Mistral LLM, optimizing prompt construction and achieving a 30% faster inference.
- Developed a context-aware Q&A chatbot that delivers strategy-focused answers and minimizes irrelevant responses.